Accessing Heritage of Nazi Persecution with Digital Means:Ethical Treatment and Inclusive Design

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Heritage of Nazi persecution (HNP) poses a particular challenge for computer-based visualisation and design. Despite well-organised and multi-faceted collections, most of which already digitised, HNP is characterised by a high degree of fragmentation and heterogeneity (Blanke et al. 2014)[1]. This gives rise to ethical considerations, which are discussed in this paper with regard to an inclusive design based on multiple perspectives and personalised experiences.

The Difficult Heritage of Nazi Persecution

The history of Nazi persecution constitutes an important reference point for democratic societies, specifically in the face of misinformation and falsification of historical facts (Verschure and Wierenga 2021: 331). For that purpose, the sources that constitute HNP need to be preserved and made accessible in accurate, responsible but nevertheless engaging ways, not only to resist forgetting but also to create new resonance (Walden 2023; Ebbrecht-Hartmann et al. 2023).

The Horizon Europe project MEMORISE (2022-2026)[2] explores how digital technology enables new ways of engaging with HNP in interactive environments, and linking contextual knowledge. However, turning HNP into the subject of digital enterprise is not without ethical and logistical-technological challenges (Makhortykh 2021; Makhortykh et al. 2023). Particularly, there have been repeated discussions about how the persecution and mass murder of Jews, Sinti and Roma, and other persecuted groups can be appropriately presented and remembered in a form that does justice to the dimension of the crimes, the fragmentary nature of the historical record, and protects the victims' dignity.

Our project addresses this challenge by enabling users to have individualized experiences through interacting with personal stories in digital storytelling environments. Subjective sources such as testimonies, letters and diaries were long ignored by historians. While they require careful handling due to their subjective nature they also offer atmospheric insight for digital tools that memory institutions cannot provide. Next to written sources, prisoner artworks are particularly central to our approach as an example for visual heritage. They humanize the victims' experiences and allow first-hand insights into multiple perspectives on Nazi persecution and thus become part of a dynamic heritage.

Principles of digitally accessible HNP

In 2006, the growing concerns for hyperrealistic heritage visualisations led to the creation of the London Charter on computer-based visualisation of cultural heritage (Denard 2009; 2012). Among

other recommendations, it encourages ensuring differentiation in evidence-based versus interpretative, heritage reconstructions, while being transparent on creative choices where these two merge. This principle coincides with the requirement of concentration camp memorial sites to preserve and utilise historical remains in their fragmentary character and carefully use reconstructions not to rebuild historic structures but to render their dimensions visible (Wagner 2019). For the digital accessibility of HNP, this means reflecting on their digital processing.

Creating 3D models and other computer-based visualisations demands careful decisions regarding historical accuracy (Wagner 2019)[3]. Applications should make it clear that 3D models cannot provide an accurate representation of historical reality, because, as it is stated in the Seville Charter, "virtual recreations…cannot systematically show lifeless cities, lonely buildings or dead landscapes, because this is a historical falsehood" (Lopez-Menchero and Grande 2011). The Seville Charter was created using the general framework of the London Charter, and specifies it for Virtual Archaeology. It emphasises interdisciplinary collaboration, ensuring an *exchange of ideas* from varying perspectives between computer specialists, historians, engineers and so forth. This principle is fundamental in HNP projects because of the sensitive nature of the topic. In the following, we demonstrate how these principles are reflected in the MEMORISE project.

Towards accessible and inclusive design

Digitisation and computer-based visualisation can help safeguard HNP by digitally documenting historical sites, artefacts, and stories, mitigating the risk of physical deterioration or loss (Knoch 2021; Wagner 2019). While staying available for physical or digital exhibitions on-site, virtual interfaces also make them accessible from a distance and without barriers. In this process, MEMORISE prioritises respect and transparency for the integrity of historical sources. This implies the possibility of switching from a "distant view" to a "close view" of HNP in its diversity and particularity (Presner 2024). MEMORISE provides both views for a collection of around 2,000 prisoner artworks. We have developed a "distant viewing" tool that allows users to explore this collection by certain objects and topics, but also stylistic details within the artwork. Filters make it possible to select thematic clusters, but also to access the individual works of art and related metadata as seen in Figure 1.

Accessing and exploring 2,000 prisoner artworks through a distant viewing tool.

Our "close viewing" approaches are illustrated with a watercolour drawing made by former Bergen-Belsen prisoner Ervin Abadi showing an exterior view of the Bergen-Belsen concentration camp. Firstly, we can use this visual testimony as the starting point for reconstructing barracks and other camp structures in 3D (Hall et al., 2024). Only original visual elements were used for these models in Figure 2 (top). Where this was not possible, the structures were deliberately kept abstract in order to distinguish them from the coloured textures, as recommended by experts from memorial sites (Wagner 2019).

In addition, access is provided by a digital reproduction of the two-dimensional drawing in its overall context, so that its adaptation into a three-dimensional model remains transparent. This is in line with the Seville Charter, which states: "it should always be possible to distinguish what is real, genuine or authentic from what is not" (Lopez-Menchero and Grande 2011).

In alignment with above outlined recommendations, the way HNP is handled in the MEMORISE project is largely based on the expertise of historians and experts, whose knowledge is used to contextualise and enrich data. In addition, the linking of various testimonies, letters, diaries, transcribed video testimonies and drawings combines historical information with experiential knowledge. Digital visualisations are grounded on expert documentation and research, displaying all documented periods in varying stages, instead of depicting an idealised image of the past. For that reason, the 3D models of prisoner paintings are not only localised in terms of time with regard to their creation, but also as a supplement to 3D models that visualise different phases in the development of the camp. Furthermore, as first-hand

visual memories they help estimate life and suffering in the camp.

Interactive 3D Model and 2D Explorer based on a painting by Ervin Abadi.

Experiential Knowledge and Inclusive Heritage

Computerisation methods can improve the readability of historical sources (Wagner 2019). To this end, we have developed an application that allows users to explore individual details and the various compositional layers of Abadi's painting in Figure 2 (bottom). Extracted objects-of-interest in multiple painting layers act as interactive entry points to zoom into the details and annotate the contents by further contextualising information. This enables a free exploration of painting elements or linear scrollytelling, an intuitive narrative technique, appealing to the habitual content presentation on mobile devices (Seyser & Zeiller, 2018). Without harming the integrity of the source, users can also hide elements of the painting in order to better understand its compositional principles and the creation of meaning.

Thus, digitised and visualised heritage can function as an entry point into multi-layered environments that connect various heritage sources. Their combination with personal stories and individual biographies strengthens cognitive-empathy and allows users to relate to a variety of perspectives on the history of Nazi persecution as well as a better understanding of the interrelation and significant differences of past and present experiences. In our project, we shift attention from the existing historical canon to less known biographies, sites and stories that are continuously overlooked or ignored. Paintings, which the Romani Bergen-Belsen survivor Ceija Stojka has created decades after her liberation, offer entry points for users of broad audience groups to engage with fragments of her testimonies and explore background information about her experiences in Bergen-Belsen and the persecution of Sinti and Roma.

This approach gives voice to marginalised experiences and diversifies the stories told about Nazi persecution as well as the personalised experiences of users and the experiential knowledge they gain from them.

Conclusion

Innovative technology can offer interactive experiences bridging past and present (De Jong 2023). Engaging with HNP in digital environments has to reflect specific ethical and representational practices and limitations that result from the unprecedented character and scale of systematic mass persecution. This makes it important to carefully reflect the relation of HNP and computer-based technologies as well as the adequacy of utilising such technologies in this particular context. A multi-layered digital (or hybrid) environment in which visitors to memorial sites can explore various reconstructed histories and historical viewpoints can help shift attention from the existing historical canon to biographies, sites, objects and stories that are less known and often overlooked or marginalised.

Notes

1. Towards the end of the war, the Nazis tried to erase the traces of their crimes. Official documents are often characterised by a bureaucratic language of camouflage, and persecution was based on the comprehensive collection and evaluation of data on the persecuted, who were reduced to mere numbers and registered in lists in the camps (Aly / Roth 2004).
2. More information: https://memorise.sdu.dk/ .
3. Considerations include whether or not to 'fill in' missing information, to 'crop' or alter objects, 'fix mistakes', and acknowledge the extent of historical reconstruction, all of which must be approached thoughtfully and transparently. For that reason, digital visualisations must not be employed as an end in itself or as a mere 'gimmick'.

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